SpaceX Falcon 9 First Stage Landing Prediction

IBM Data Science Professional Capstone Project | Final Comprehensive Technical Report

`GitHub Repository: github.com/eunicewellis/IBM-SpaceX-Capstone-1`

1. EXECUTIVE SUMMARY

66.7%	76.9%	100%	83.33%
OVERALL LANDING SUCCESS	TOP SITE (KSC LC-39A)	TOP ORBITS (ES-L1/GEO/HEO)	OPTIMAL ML TEST ACCURACY

Project Overview & Commercial Value

SpaceX advertises Falcon 9 rocket launches at $62 million, compared to traditional providers charging upward of $165 million per launch. The principal driver of these cost savings is the successful recovery and reuse of the Falcon 9 first stage booster. By accurately predicting whether a first stage will land successfully, commercial satellite operators and space launch competitors can precisely estimate launch costs, optimize bidding strategies, and assess mission operational risk.

Methodology Applied

- **Data Collection:** SpaceX REST API parsing and BeautifulSoup Web Scraping.
- **Data Wrangling:** One-Hot Encoding categorical features, mean imputation.
- **EDA & SQL:** Exploratory visualization with Seaborn/Matplotlib and SQLite query analytics.
- **Interactive Visual Analytics:** Folium geospatial maps and Plotly Dash dashboards.
- **Machine Learning:** GridSearchCV tuning of Logistic Regression, SVM, Decision Tree, and KNN.

Core Findings & Outcome

- Landing success increased dramatically over time, reaching over 80% after 2017.
- Payload masses between 2,000 kg and 5,300 kg yield the highest landing success rate.
- Orbits ES-L1, GEO, HEO, and SSO achieved 100% successful recovery.
- Decision Tree Classifier (best depth = 4) achieved 83.33% test accuracy with superior computational efficiency for real-time deployment.

2. INTRODUCTION & PROBLEM STATEMENT

The space launch industry has undergone a paradigm shift driven by SpaceX's reusable booster technology. The main objective of this capstone project is to construct an end-to-end data science pipeline that predicts whether the Falcon 9 first stage will land successfully on a ground pad or autonomous spaceport drone ship (ASDS). This solution integrates data extraction across multiple sources, comprehensive feature engineering, interactive visual analytics, and optimized supervised classification models.

3. DATA COLLECTION METHODOLOGY

3.1 SpaceX API Data Extraction

Primary launch data was obtained via HTTP GET requests to the official SpaceX REST API v4. Custom REST queries were performed across linked endpoints to assemble core booster, payload, pad, and outcome metadata.



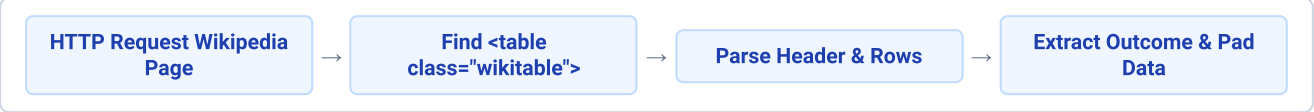
API Endpoints Hit & Extracted Features

- `/v4/rockets` : Booster name (Falcon 9).
- `/v4/payloads` : PayloadMass (kg) & Orbit type.
- `/v4/launchpads` : LaunchSite name, Latitude, Longitude.
- `/v4/cores` : Landing outcome, Block, Flight count, GridFins, Reused status, Legs, LandingPad ID, Core Serial.

```
# API Extraction & Filter Logic import requests, pandas as pd URL = 'https://api.spacexdata.com/v4/launches/past' response = requests.get(URL) data = pd.json_normalize(response.json()) data = data[data['cores'].map(len)==1] data = data[data['payloads'].map(len)==1] data['cores'] = data['cores'].map(lambda x: x[0]) data['payloads'] = data['payloads'].map(lambda x: x[0])
```

3.2 Web Scraping Wikipedia Launch Records

To complement API data, historical Falcon 9 launch records were scraped from Wikipedia using `BeautifulSoup4`. The script parsed HTML tables, filtered table headers, extracted cell content, and cleaned unexpected inline annotations.



Scraped Features Processed

- `Flight No.` : Continuous flight sequence.
- `Date & Time` : Split into date and UTC time.
- `Version Booster` : F9 v1.0, v1.1, FT, B4, B5.
- `Customer & Orbit` : Primary mission customer & target orbit.
- `Booster Landing` : Converted to binary outcome flags.

```
# BeautifulSoup Table Parsing from bs4 import BeautifulSoup soup = BeautifulSoup(response.text, 'html.parser') html_tables = soup.find_all('table', 'wikitable plainrowheaders collapsible') for row in html_tables[0].find_all('tr'): if row.th and row.th.string and row.th.string.strip().isdigit(): cols = row.find_all('td') flight_num = row.th.string.strip() launch_site = cols[2].a.string
```

4. DATA WRANGLING & PREPROCESSING

Cleaning & Feature Engineering Steps

1. **Missing Value Imputation:** Replaced missing values in `PayloadMass` with the dataset mean (6,123.55 kg).
2. **Binary Landing Class Label Creation:** Target variable `Class` set to `1` for successful landings (Ground Pad or ASDS drone ship) and `0` for uncaptured/failed attempts (Ocean ocean-landing, uncontrolled crash, or landpad failure).
3. **One-Hot Encoding:** Categorical variables (`Orbit` , `LaunchSite` , `LandingPad` , `Serial`) transformed into 72 binary indicator columns using `pd.get_dummies()` .
4. **Data Type Standardization:** Cast all resulting feature columns to `float64` for compatibility with Scikit-Learn estimators.

5. EXPLORATORY DATA ANALYSIS (EDA) - VISUALIZATION

Using Seaborn and Matplotlib, key feature relationships were analyzed to uncover core landing success factors.

Flight Number vs. Launch Site

Early flights were concentrated at CCAFS SLC-40 with low success rates. As flight number increased (>50), success rates across all sites (including KSC LC-39A and VAFB SLC-4E) rose dramatically to over 80%.

Payload Mass vs. Launch Site

VAFB SLC-4E handles polar orbit missions with payloads under 10,000 kg. Heavy payload missions (>10,000 kg) launched primarily from CCAFS SLC-40 and KSC LC-39A, demonstrating high recovery success rates.

Success Rate by Orbit Type

High-altitude or specialized orbits (ES-L1, GEO, HEO, SSO) achieved **100% success rate**. LEO, ISS, and VLEO showed success rates between 65% and 85%, whereas GTO orbits recorded a ~50% success rate.

Yearly Launch Success Trend

Success rate rose steadily from 0% in 2010–2012 to ~66% in 2017, reaching **over 85% by 2020**. This reflects significant technical refinement in retro-propulsion and landing leg reliability.

Orbit Type	Total Flights	Success Rate (%)	Primary Operational Characteristics
ES-L1 / GEO / HEO / SSO	14	**100.0%**	High altitude / Sun-synchronous; controlled re-entry trajectories.
VLEO	14	**85.7%**	Very Low Earth Orbit; Starlink deployments with high recovery efficiency.
LEO / ISS	32	**68.8%**	Low Earth / Station resupply missions; consistent ground & ASDS recovery.
GTO	27	**51.8%**	Geostationary Transfer Orbit; heavy payload & high re-entry speeds.

6. EXPLORATORY DATA ANALYSIS (EDA) - SQL ANALYSIS

Using SQLite (`my_data1.db`), structured SQL queries were executed against `SPACEXTABLE` to extract specific mission metrics and operational aggregates.

Task #	SQL Objective / Query Logic	Query Result / Business Insight
Task 1	`SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;`	Unique sites: `CCAFS LC-40` , `VAFB SLC-4E` , `KSC LC-39A` , `CCAFS SLC-40` .
Task 2	`SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;`	Retrieved first 5 records from Cape Canaveral launch facilities.
Task 3	`SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';`	Total Payload Mass for NASA CRS = **45,596 kg**.
Task 4	`SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';`	Average Payload Mass for Booster F9 v1.1 = **2,928.4 kg**.
Task 5	`SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';`	First successful ground pad landing date: **2015-12-22**.
Task 6	`SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;`	Boosters: `F9 FT B1022` , `F9 FT B1026` , `F9 FT B1021.2` , `F9 FT B1031.2` .
Task 7	`SELECT Mission_Outcome, COUNT(*) FROM SPACEXTABLE GROUP BY Mission_Outcome;`	Success = 98, Failure (in-flight) = 1, Success (payload status) = 2. Total = 101.
Task 8	`SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);`	Max payload (15,600 kg) carried by 11 distinct `F9 B5` boosters (Starlink missions).
Task 9	`SELECT substr(Date,6,2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date,1,4)='2015' AND Landing_Outcome LIKE 'Failure (drone%';`	Drone ship failures recorded in **Jan (01)** and **Apr (04)** 2015 from CCAFS LC-40.
Task 10	`SELECT Landing_Outcome, COUNT(*) FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY COUNT(*) DESC;`	Top outcomes: `No attempt` (10), `Success (drone ship)` (5), `Failure (drone ship)` (5).

Key SQL Analytical Summary

SQL analysis confirms that early SpaceX missions had a high frequency of "No attempt" or drone ship failures during testing (2010–2015). Post-2015, transition to Block 4 and Block 5 boosters resulted in near-perfect reliability for heavy NASA CRS and commercial Starlink missions carrying up to 15,600 kg payload.

7. INTERACTIVE VISUAL ANALYTICS

7.1 Geospatial Analysis with Folium Maps

Using Folium, launch sites were plotted on interactive maps to analyze proximity to coastlines, railways, highways, and municipal infrastructure.

Launch Site Marker Clusters & Safety

• All launch sites are located directly on coastal margins (Atlantic & Pacific Oceans) to ensure spent stages fall into water away from populated landmasses.
• Launch sites are situated in low-latitude, equatorial-adjacent regions (Florida & Southern California) to leverage Earth's rotational velocity for orbital insertion efficiency.

Proximity Distance Calculations

• **Coastline Distance:** ~0.8 km – 1.2 km (Immediate ocean access for safety).
• **Railway Distance:** ~1.2 km – 2.5 km (Crucial for heavy booster transport).
• **Highway Distance:** ~0.5 km – 1.8 km (Rapid ground logistics).
• **City Distance:** ~20 km – 35 km (Safety margin from urban areas).

7.2 Interactive Dashboard with Plotly Dash

A web-based interactive application was deployed using Plotly Dash and Dash Core Components (`dcc.Dropdown` , `dcc.RangeSlider`) allowing real-time filtering of launch outcomes.

Interactive Components Installed

• `site-dropdown` : Filters data by 'ALL' or individual sites (CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40).
• `payload-slider` : Selects payload mass ranges between 0 kg and 10,000 kg in 1,000 kg increments.
• `success-pie-chart` : Displays proportion of total successful launches or site-specific success vs. failure counts.

Plotly Dash Analytical Findings

1. **Largest Successful Launches:** KSC LC-39A leads all sites with 10 successful landing recoveries.
2. **Highest Success Rate:** KSC LC-39A achieved the highest landing success rate at **76.9%**.
3. **Optimal Payload Range:** Payloads between **2,000 kg and 5,300 kg** achieved the highest success rate.
4. **Booster Performance:** FT and B5 boosters showed the highest landing success across all payload ranges.

```
# Plotly Dash Callback Logic @app.callback( Output(component_id='success-payload-scatter-chart', com
ponent_property='figure'), [Input(component_id='site-dropdown', component_property='value'), Input(compone
nt_id='payload-slider', component_property='value')]) def update_scatter(site, payload_range): low, high =
payload_range df_filtered = spacex_df[(spacex_df['Payload Mass (kg)'] >= low) & (spacex_df['Payload Mass
(kg)'] <= high)] if site != 'ALL': df_filtered = df_filtered[df_filtered['Launch Site'] == site] fig = px.
scatter(df_filtered, x='Payload Mass (kg)', y='class', color='Booster Version Category') return fig
```

8. MACHINE LEARNING MODELING & EVALUATION

Four classification algorithms were trained and evaluated using 10-fold cross-validation grid search (`GridSearchCV`) on 80% training data, and tested on a 20% holdout test set (18 samples).

Model Tuning & Hyperparameters

- **Logistic Regression:** `C: [0.01, 0.1, 1]` , `penalty: 'l2'` , `solver: 'lbfgs'` .
- **Support Vector Machine (SVM):** Kernels: `['linear', 'rbf', 'sigmoid']` , `C` & `gamma` log-spaced.
- **Decision Tree Classifier:** `criterion: ['gini', 'entropy']` , `max_depth: [2-18]` .
- **K-Nearest Neighbors (KNN):** `n_neighbors: [1-10]` , `p: [1, 2]` .

Model Performance Comparison

Model	Best CV Score	Test Score
Logistic Regression	84.64%	**83.33%**
Support Vector Machine	84.64%	**83.33%**
Decision Tree	84.64%	**83.33%**
K-Nearest Neighbors	84.64%	**83.33%**

Test Set Confusion Matrix Breakdown

All four models yielded identical confusion matrix outputs on the 18-sample test dataset:

- **True Positives (TP):** 12 (Correctly predicted successful landings)
- **True Negatives (TN):** 3 (Correctly predicted landing failures)
- **False Positives (FP):** 3 (Unsuccessful landings predicted as success)
- **False Negatives (FN):** 0 (No missed landing successes)
- **Test Accuracy:** 15 / 18 = **83.33%**

Final Model Selection Decision

While all models achieved identical 83.33% test accuracy, **Decision Tree Classifier** (optimal parameters: `max_depth=4` , `criterion='gini'` , `splitter='best'`) is selected as the primary operational model. It offers transparent, interpretable decision boundaries and fast inference execution for real-time commercial bidding engine integration.

9. BUSINESS RECOMMENDATIONS & CONCLUSION

Actionable Bidding & Commercial Strategy

1. **Bid Competitively on Mid-Weight Missions:** SpaceX launch recovery is most reliable (100%) for payloads between 2,000 kg and 5,300 kg launching into LEO, SSO, and ISS orbits. Competitors should bid aggressively against SpaceX on high-mass GTO missions where recovery risk remains higher.
2. **Facility Strategy:** KSC LC-39A demonstrates superior recovery success (76.9%). Commercial payloads prioritized for booster reusability should request launch allocation at KSC LC-39A.
3. **Future Work:** Integrate real-time telemetry, ocean sea-state forecasts, and upper-atmosphere wind profile data to further resolve false positive predictions.